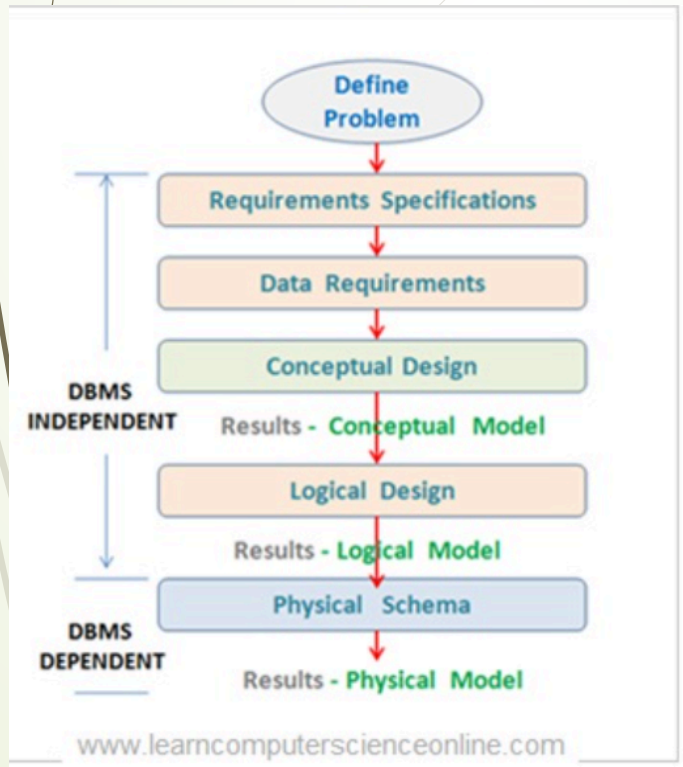
DATABASE MANAGEMENT SYSTEMS

2- Design and ER Model

Dr. Fatih KAYAALP

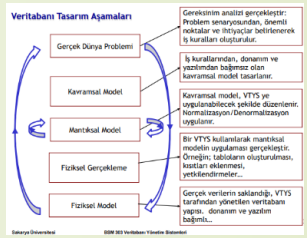
WPS Office ile Düzenle

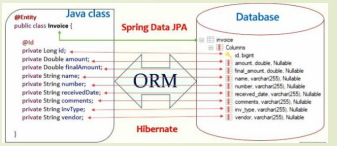
DB Modeling Stages

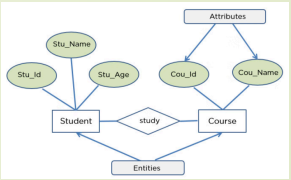


1. Define Database design Objectives
2. Database designers collects the user database requirements by elicitation .
3. Database designers produce conceptual data model then expanded to Logical data model then expanded to physical data model .
4. Database designers construct logical design of the database with tables , Defining Relations , DB Constrains by
5. Database designers then Refine & Normalize the database to minimize the data redundancy
6. Physical implementation of database is managed by the DBMS .

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Databases are modelled by using one of the followings:
- ORM (Object Role Modeling),
- ER (Entity-Relationship Model)
- UML (Unified Modeling Language)



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Business Rules Examples

You are asked to design a database for a bookstore management system according to the following business rules:

- The bookstore controls several authors. Each author has a unique ID, name, National Identity Card, date of birth and city.
- Each author may write many books. Each book must be written by one author. Each book is characterized by its unique code, title, year, price and the number of copies the bookstore contains for this book.
- The bookstore keeps track of different customers. Each customer has a unique ID, name and email address.
- A customer may generate many invoices while one invoice must be associated with one customer.
- An invoice may have many books sold while one book may appear on many invoices. An invoice is identified by a unique invoice number and each invoice has an invoice date. The number of copies sold for each book in an invoice should be recorded.

Business Rules:

1. A student can only be in one department, but one department has many students.
2. Each book belongs to a category (e.g. Fiction, History, Biography).
3. Several books can be in one category.
4. Each book can have many copies in the library, although some books have only one copy. Each copy of the book is identified by a serial number to track who borrows it.
5. One student can borrow one or more books at a time from the library.
6. Once a student returns a book, the same book can be loaned to other students.

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Business Rules

- An owner may have more than one pet.
- A pet will only have one owner; however, an owner can have many associated contacts.
- An owner may have more than one phone number.
- There are different fees for different sized animals.
- Each employee will have a title (Veterinarian, Groomer, Staff, etc) but may work at more than one operation (ex. Grooming and Boarding).
- There will be separate charges (bills) for each operation. (If a pet is given treatment at the vet and is also groomed on the same day, there will be two separate charges)
- There will only be one Veterinarian associated with an invoice for Veterinary Services.

Car Rental Scenario Business Rules

- Each customer can rent one or more vehicles.
- Each rental must be for one customer and one vehicle.
- The same vehicle cannot be rented to two different customers between the same dates.
- Each vehicle's license plate number must be unique.
- Rental period must be at least 1 day.
- The model and brand of each vehicle must be recorded.
- Every employee can carry out the rental transactions.
- Each vehicle must belong to a certain class (economy, SUV, luxury, etc.).

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Hospital Scenario Business Rules

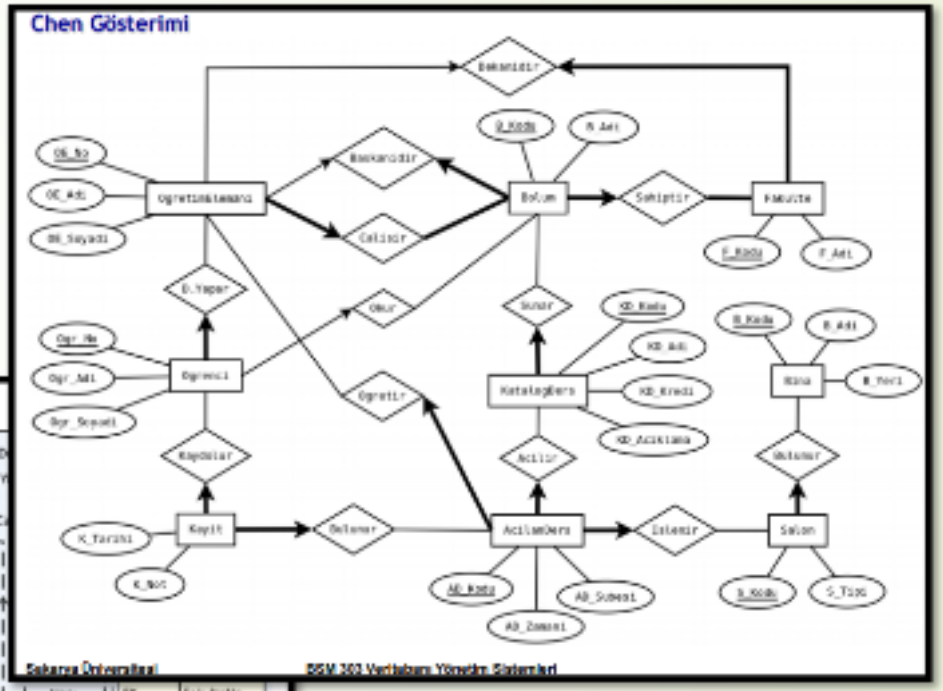
- Each patient may have one or more appointments.
- A doctor can only make one appointment at a time.
- Every doctor should have a specialty.
- Each prescription must belong to a patient and a doctor.
- No two medications have the same name and dosage combination.
- Each prescription may contain one or more medications.
- Two patients cannot be registered with the same TR ID number.

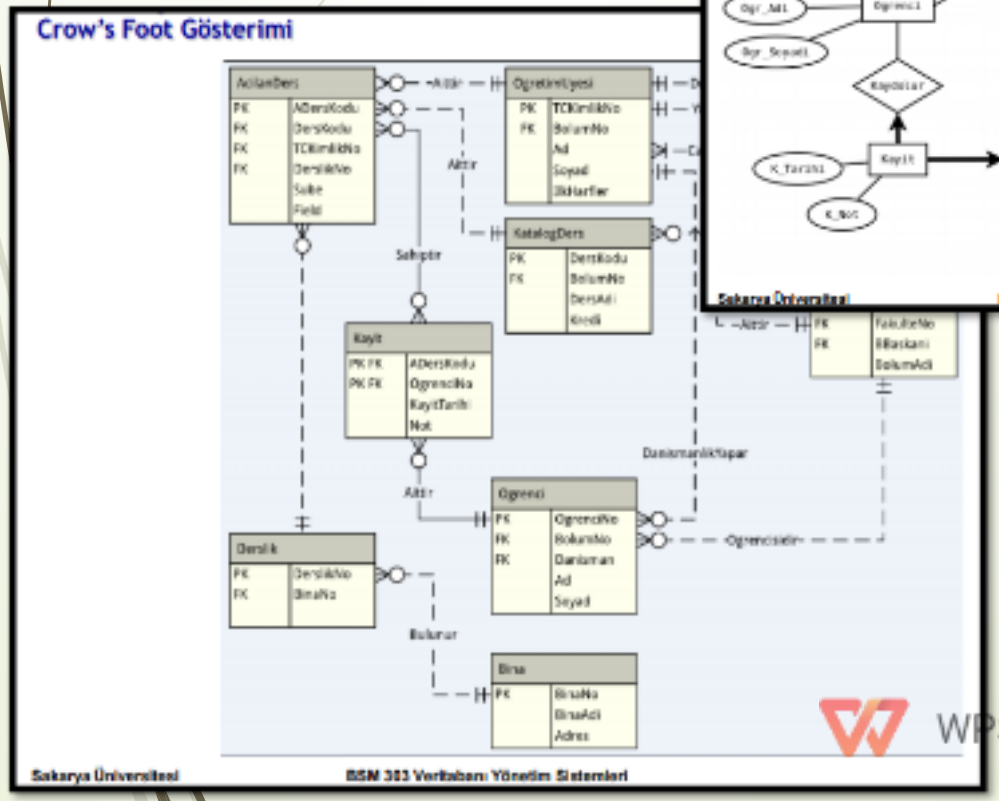
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Entity Relationship Model (ER)

- The most widely used method for conceptual design is ER model.
- The entity-relationship (ER) model, which is a graphical representation of the relational model for data design and frequently used.
- It was proposed by Peter Chen in 1976.
- Its use has become quite widespread because it is complementary to the relational data model.
- Commonly used two alternatives i) Chen notation and ii) Crow's Foot notation.

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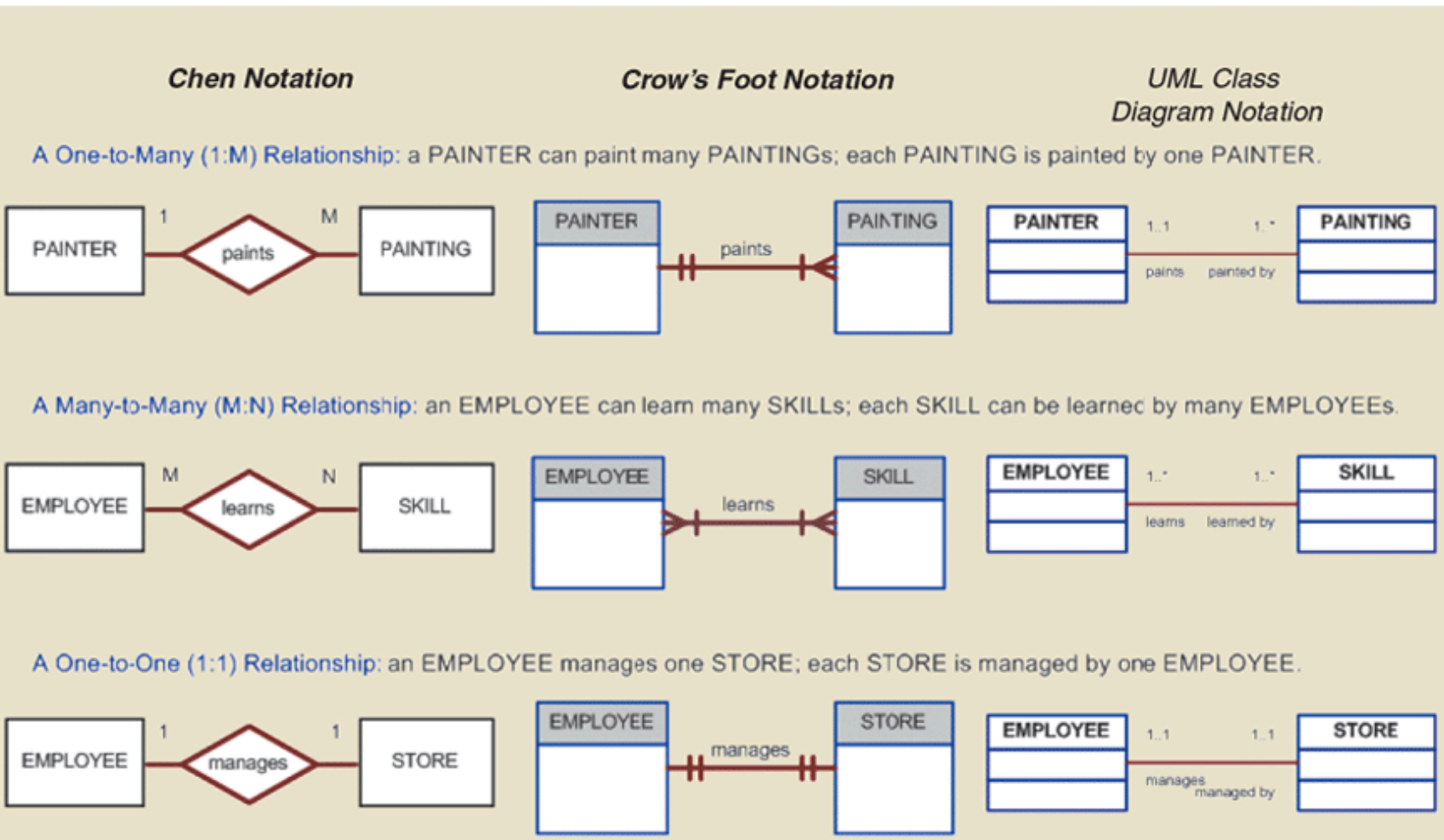
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zero or more [b4]

1 or more [b5]

1 and only 1 (exactly 1) [b6]

zero or 1 [b7]



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http://www.csc.villanova.edu/~enwafor/database/lectures/lecture_3.pdf

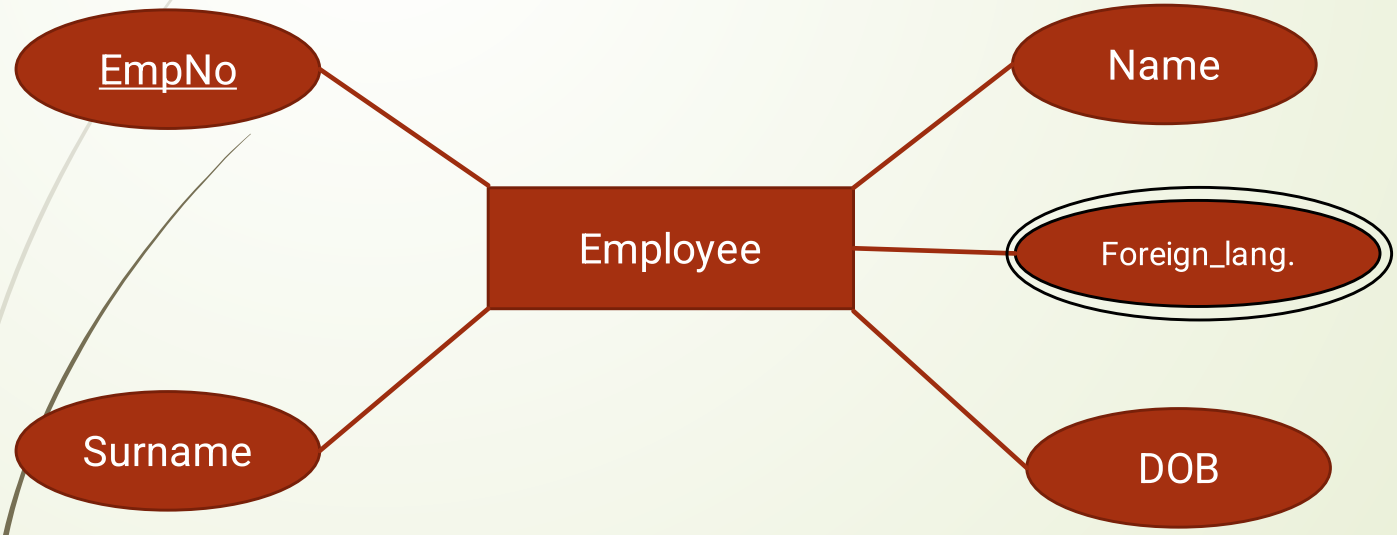
Entity Relationship Model (ER)



Concept	Explanation
Entity	The model's basic element. A single unique object in the real world that is being focused. Examples of an entity are a single person, a single product, or a single organization. Entity is shown as a rectangle in the model and the name of the entity is written inside it. «Table» is the equivalent of entity in relational model.
Attribute	Each characteristic of the entities is called attribute. Student's name, number, class... Oval shape drawn and linked to the entity in ER Model and the name of the attribute (columns of the table) is written inside. If the value of an attribute is always unique for each record, it is called «key attibute» (primary key) and is shown as underlined attribute in oval.
Multi-valued attribute	An attribute which can take more than one value. For example, an employee may know more than one foreign language. Multi-valued attributes are represented by a double-lined oval in the ER Model.
Derived attribute	???

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Entity Relationship Model



An Employee entity illustration

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https://beginnersbook.com/2015/04/er-model-in-dbms/#:~:text=Multivalued%20attribute%3A,phone%20number%20attribute%20is%20multivalued.

Entity Relationship Model

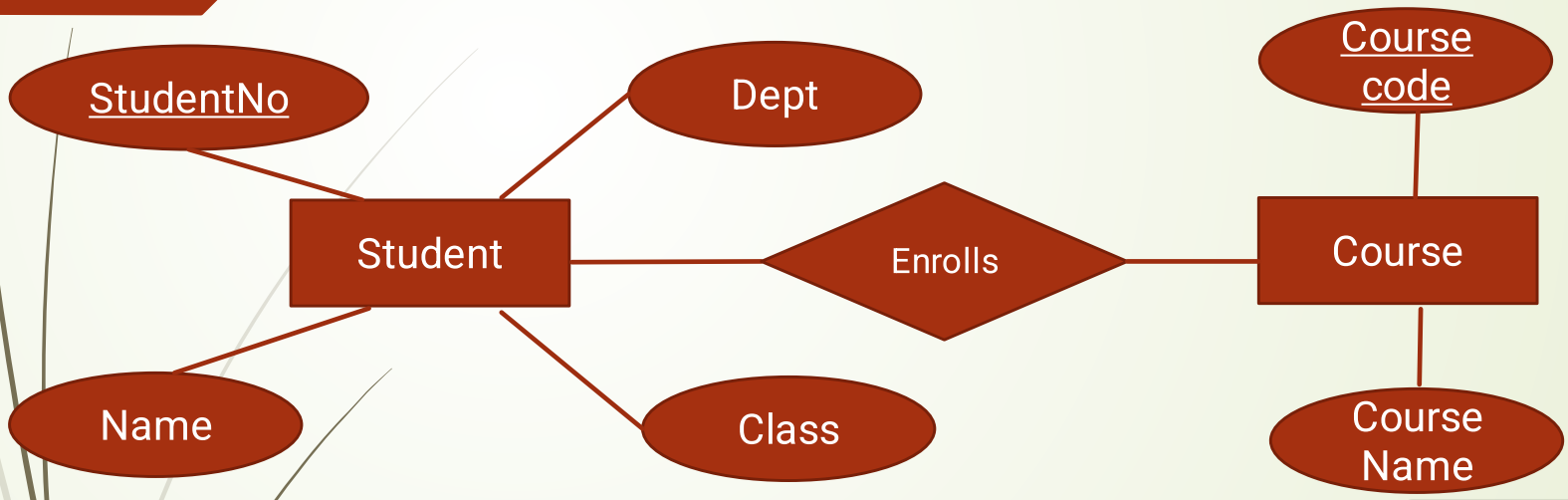
-List the attributes and draw ER diagram of **Book** and **Car** entities

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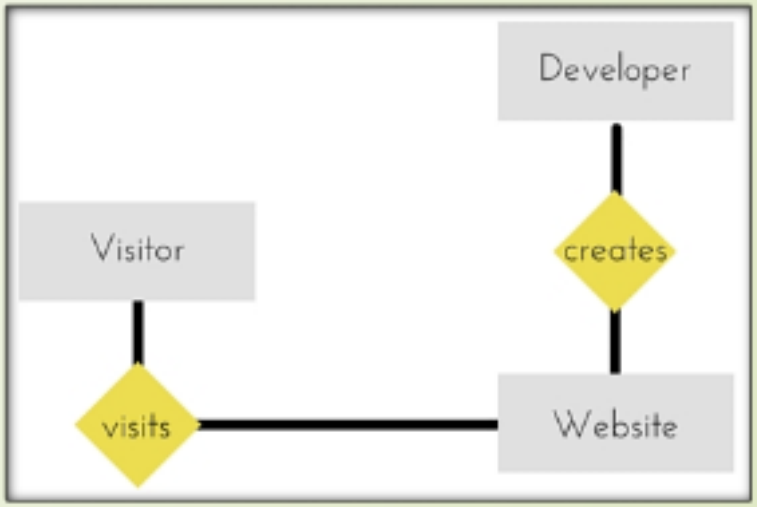
Entity Relationship Model

Concept	Explanation
Domain	The range of values that any attributes can take The domain is not shown in the ER Model.
Relationship	It expresses relationships between entities. For example, although students and courses are separate entities, there is a course-taking relationship between them because students are required to take courses. Relationships are represented by a diamond shape within the ER Model, and the name of the relationship is written inside it. The diamond shape is connected to the entities it is associated with by a straight line.

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Student-Course ER Model





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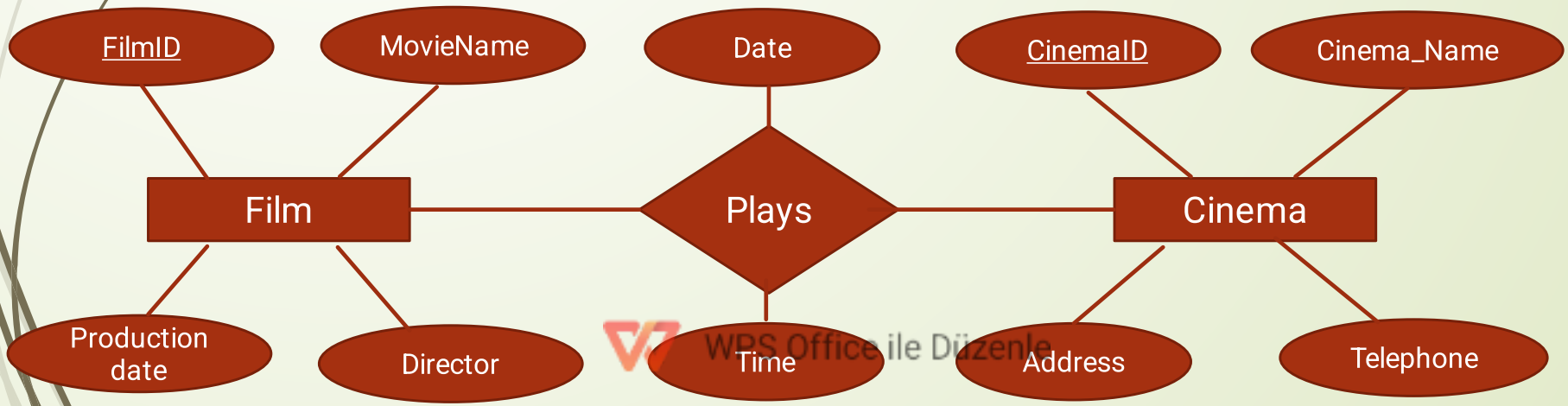
Personnel-Department ER Model

-Draw **Author** entity with attributes

-Please indicate the relationship with **Book** entity

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Concept	Explanation
Descriptive Attribute	Qualities may arise as a result of relationships formed between entities. These are called “descriptive qualities.” It is represented by an oval in the ER model and connected to the relationship by a straight line. If no relationship is created, the defining attribute never comes into existence.



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Types of Relationships

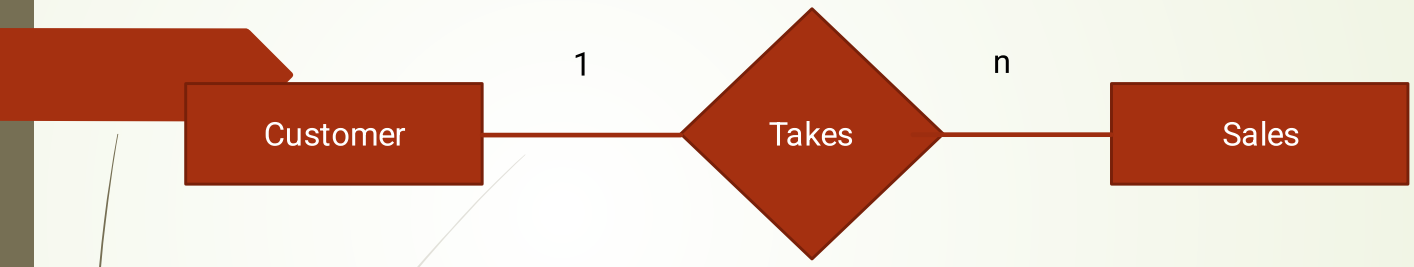
- 1-1 Relationship (One-To-One)
  - A situation where a record in one of two record sets matches only one record on the other side.

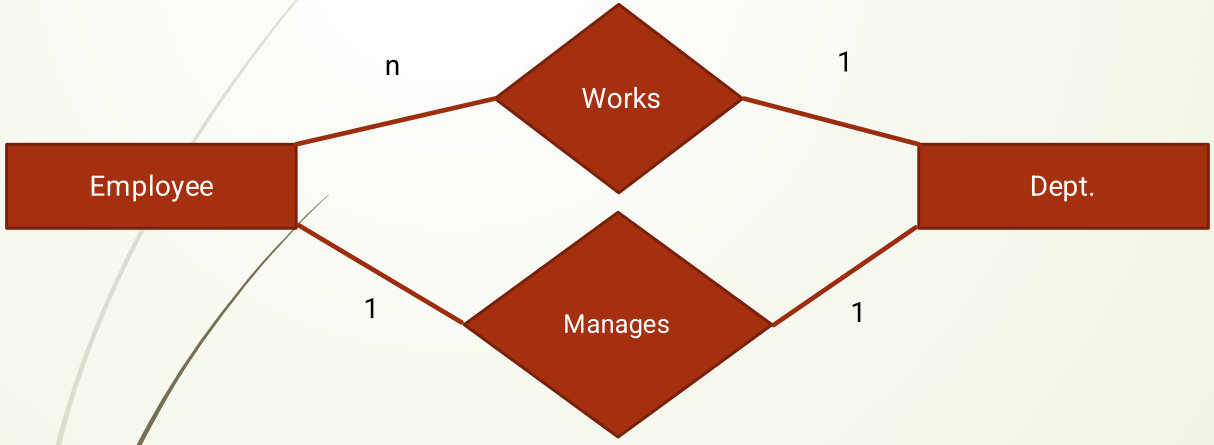
  For example, a man married to only one woman.
- 1-n (One-To-Many/Many-To-One, 1-*)
  - A situation where a record in one of two entities matches many records on the other side.

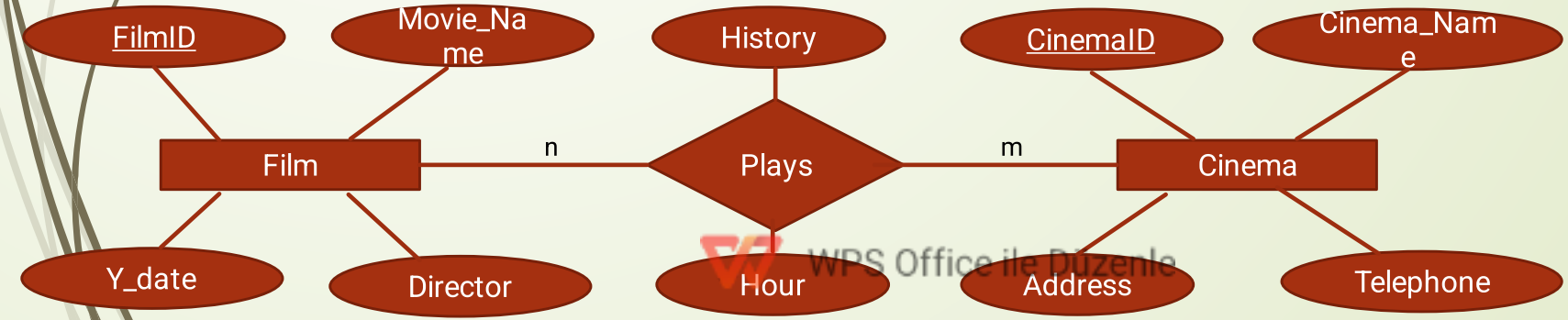
  For example, a patient can be examined multiple times.
- Many-To-Many (n-m, *-*)
  - A situation where records on both sides of two entities match many records on the other side.

  For example, while a movie is playing in different cinemas at the same time, other movies are also playing in the same cinema.

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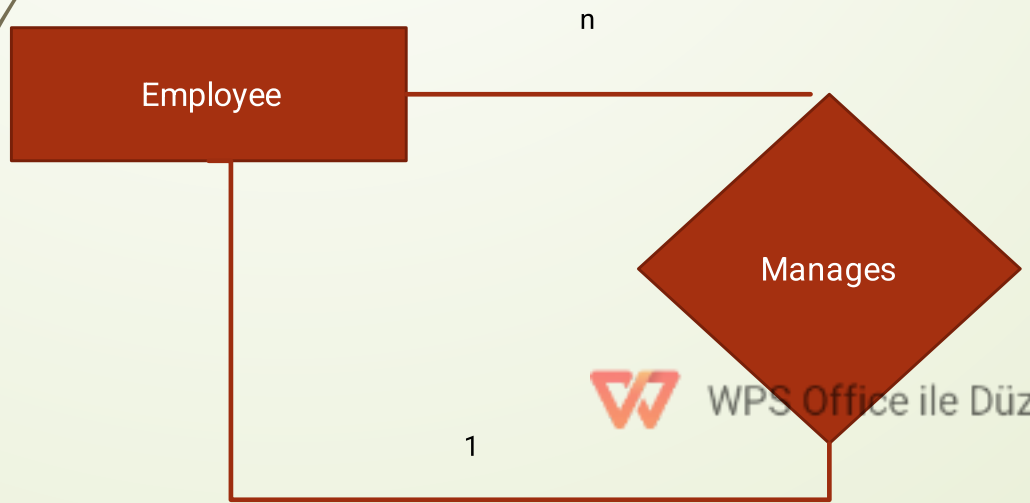




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- **Recursive Relationship:** Relationships are often created between different sets of entities. However, in some cases, a single set of entities can establish a relationship with itself.

For example, a person can be both an employee and a manager. This means that more than one person can manage a person, and a person can manage more than one person.



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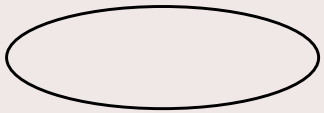
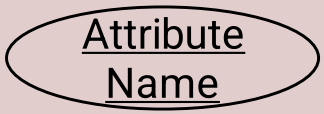
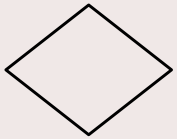
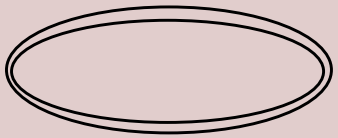
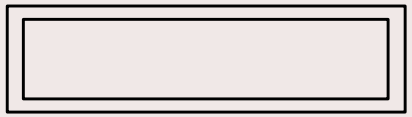
- **Weak Entity Set:** If an entity set does not have a key attribute, it is called a weak entity set. Weak entity sets are used in conjunction with strong entity sets.

That is, weak entity sets have an existence dependency on strong entity sets. Weak entity sets are represented by a double-lined rectangle in the ER Model.

Since a faculty cannot exist without a university in the university-faculty relationship and since there may be faculties in other universities with the same faculty name, the faculty entity set is a weak entity.



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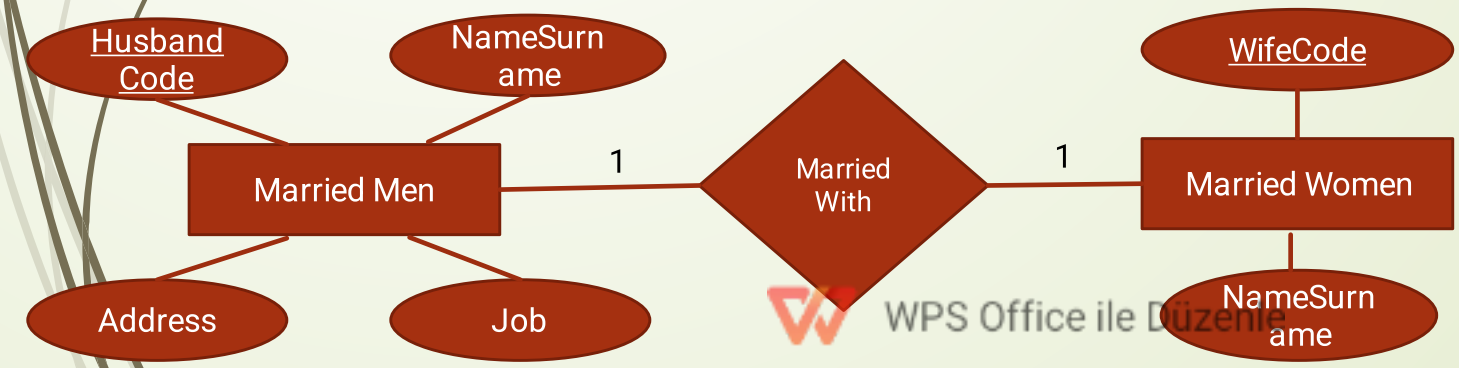
Symbol	Explanation
	Entity
	Attribute
	Key Attribute
	Relationship
	Multi-valued Attribute
	Weak Entity

WPS Office ile Düzenle

Converting the ER Model to Tables

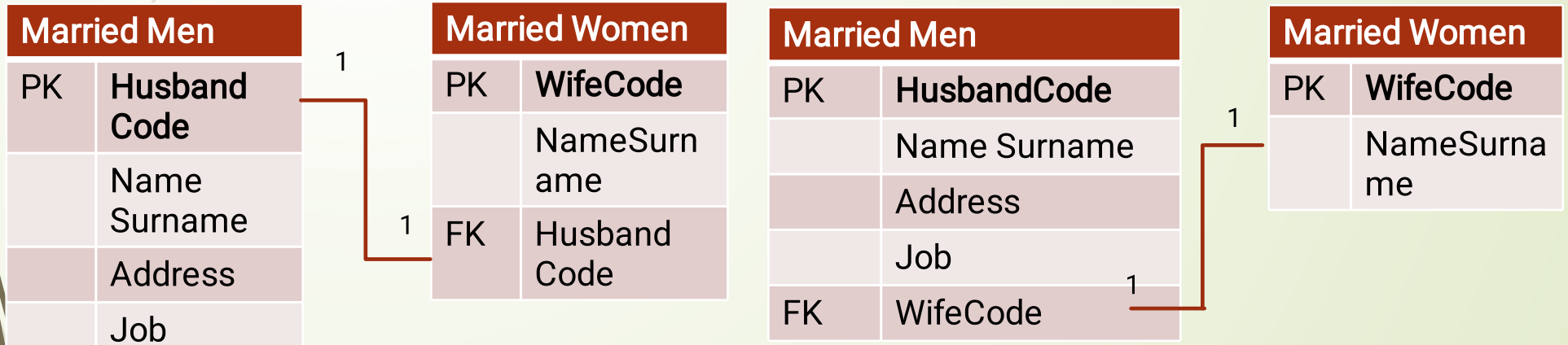
Converting One-to-One (1-1) Relationships to Tables (Database Schema)

- Entity sets are converted into tables.
- Attributes are converted to table columns.
- Primary key of one of the tables, is determined as the foreign key in the other table. Which one will be primary and which will be foreign is decided based on the table's content.
- If the relationship has descriptive attributes, the descriptive attributes are added as columns to the table that the foreign key set.



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1st Way	2nd Way

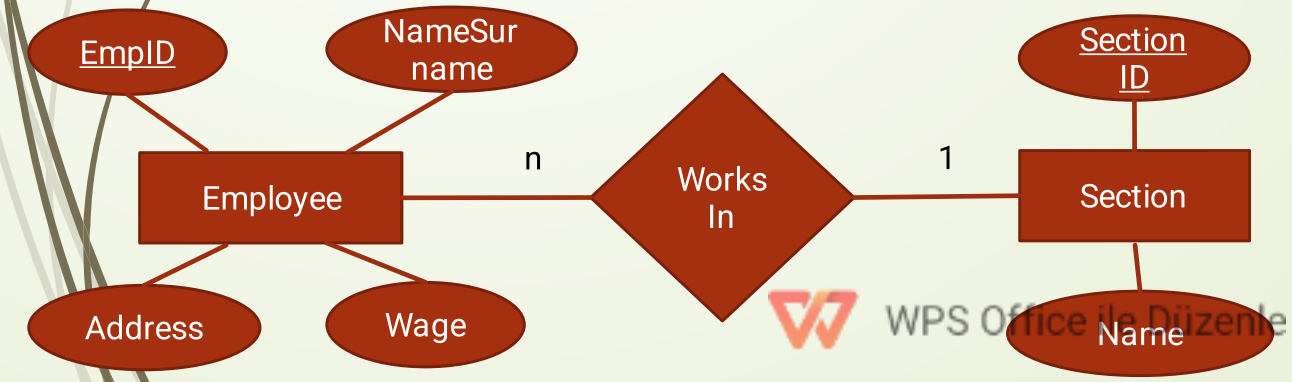


- The foreign key (FK) field must be set as unique.

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Converting One-to-Many (1-n/n-1) Relationships to Tables

- Entity sets are converted into tables.
- Attributes are converted to table columns.
- The primary key column of the table on relation side **1** is added as a foreign key on the next table.
- If the relationships created in the model contain descriptive attributes, the descriptive attributes are added as columns to the table on the relation's **n** side.



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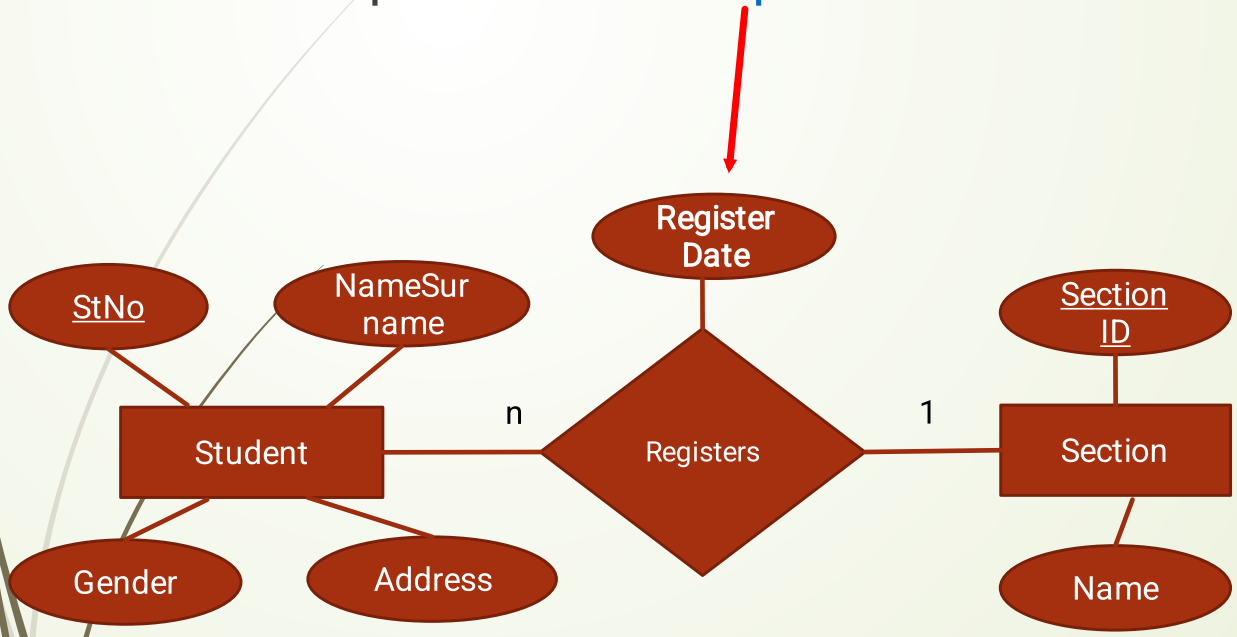
Staff Table	
PK	**PerID**
	Name Surname
	Address
	Wage
FK	Section ID

1

n

Section Table	
PK	**Section ID**
	Ordinary

- Example of a «Descriptive Attribute» in a Relationship



Student Table	
PK	**StNo**
	Name Surname
	Address
	Gender
FK	Section ID
	Register

1

n

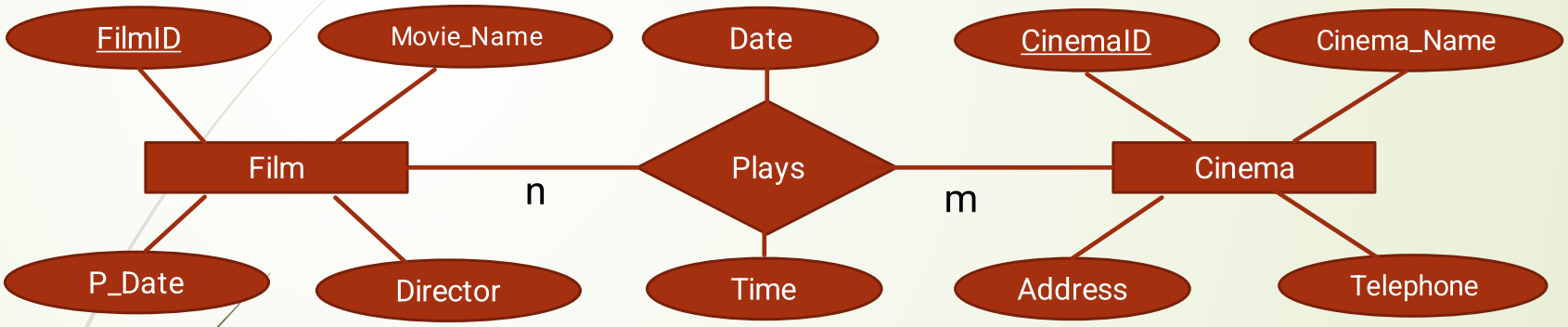
Section Table	
PK	**Section ID**
	Ordinary

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Converting Many-To-Many Relationships to Tables

- Entity sets are converted into tables.
- A table is created with the name of the relationship.
- Attributes are converted to table columns. Descriptive attributes are inserted as columns to the table created with the name of the relationship.
- The primary keys of the tables forming the relationship are added as foreign keys to the table forming the relationship.
- The primary key of the table created from the relationship consists of the combination of the foreign keys created. If the primary key created in this way does not meet the requirements, a new column is added and made the primary key.

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Movie Table	
PK	**FilmID**
	Movie_Name
	P_Date
	Director

1 — n

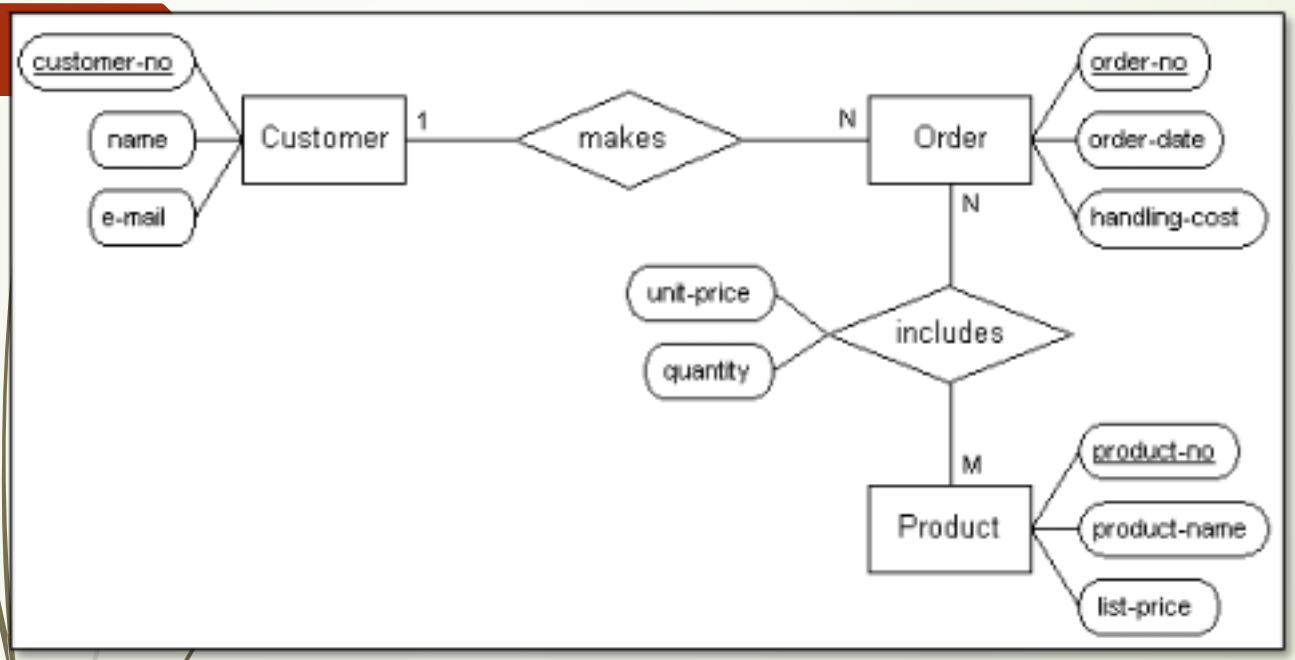
Playing Table	
PK	**PlayerID**
FK	**FilmID**
FK	**CinemaID**
	Date
	Time

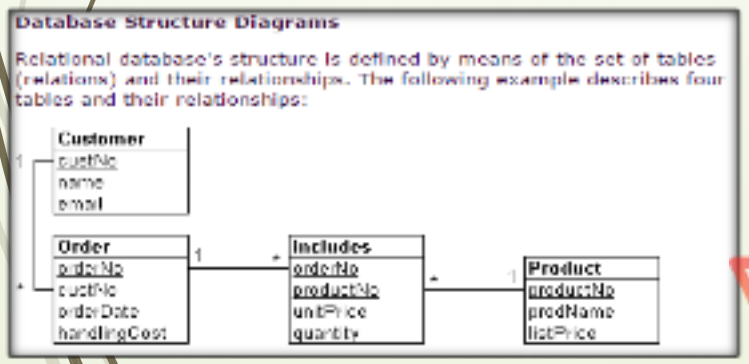
1 — n

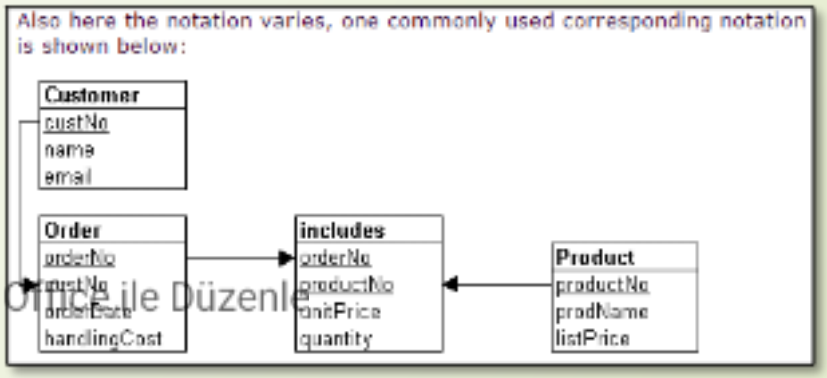
Cinema Table	
PK	**CinemaID**
	Cinema_Name
	Address
	Telephone

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Exercises-1

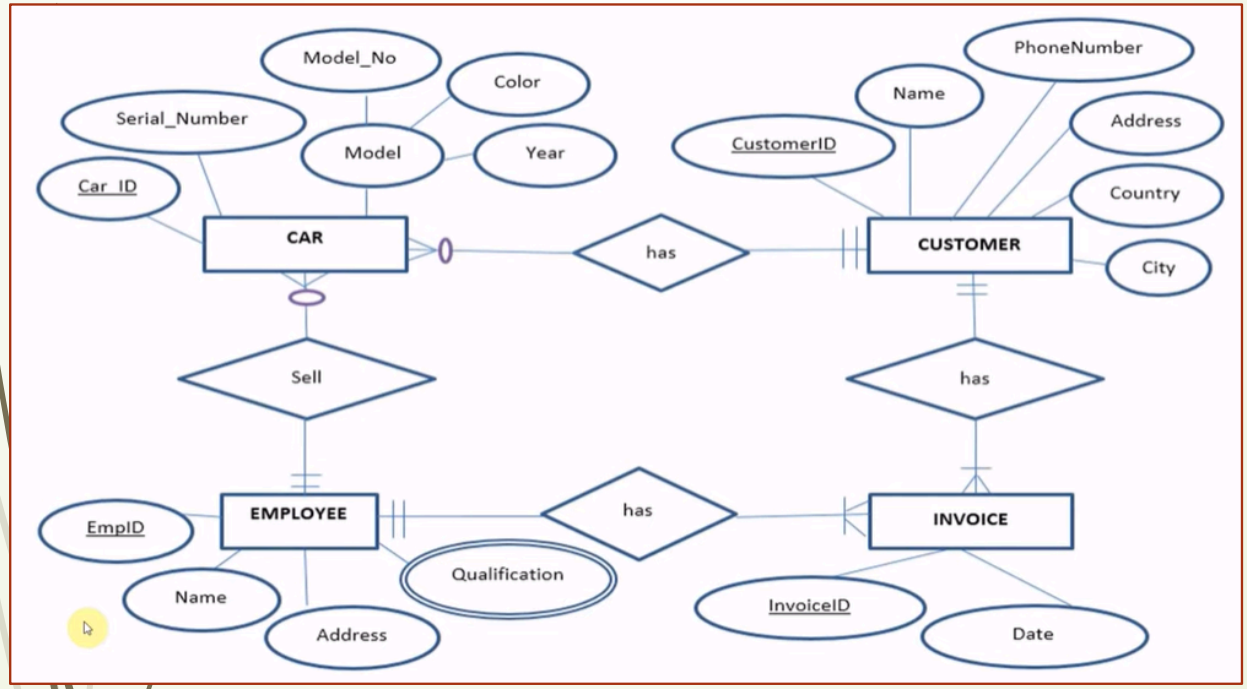






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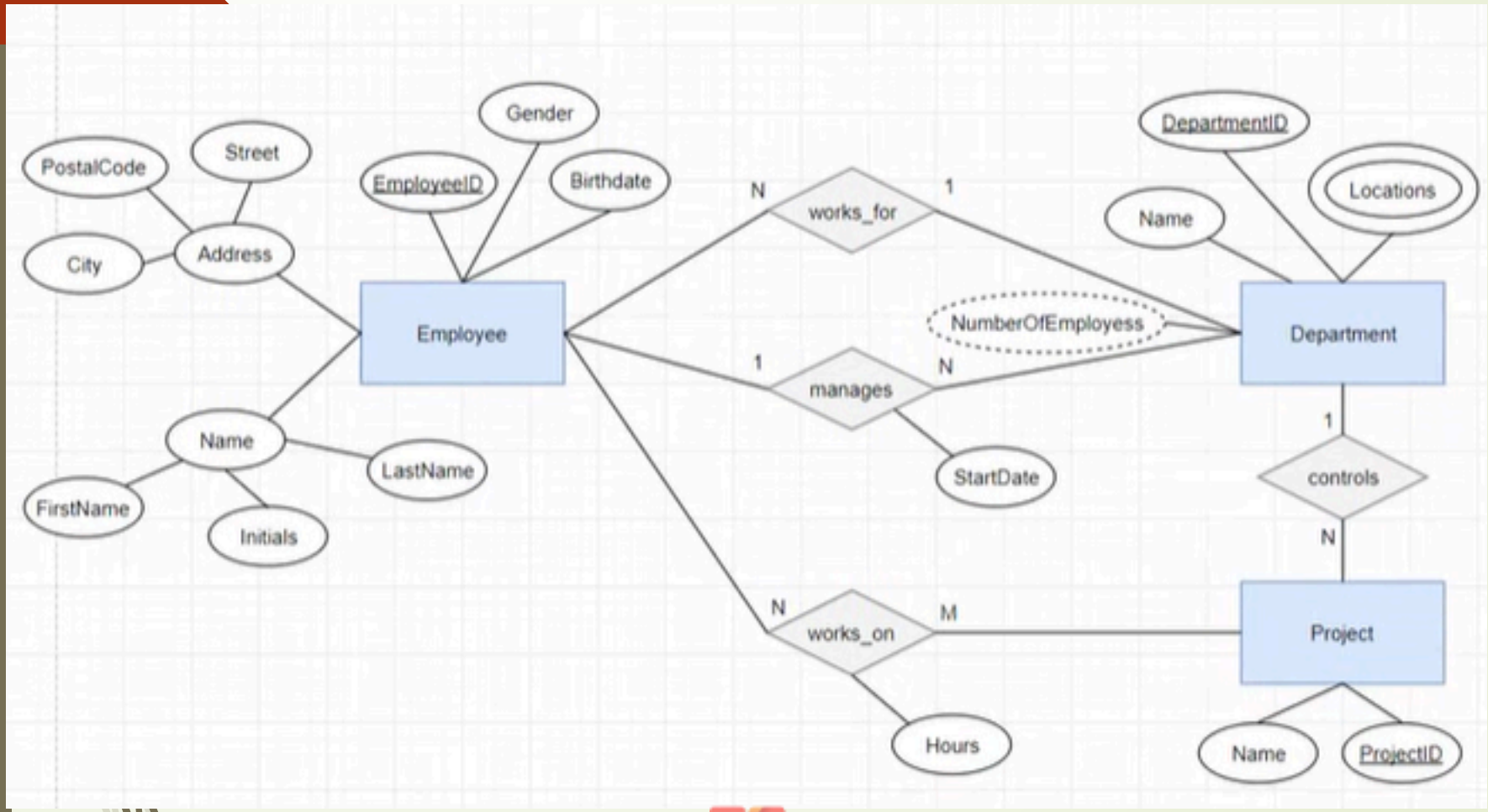
Exercises-2



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https://www.youtube.com/watch?v=OwdFzygGZqk

Exercises-3

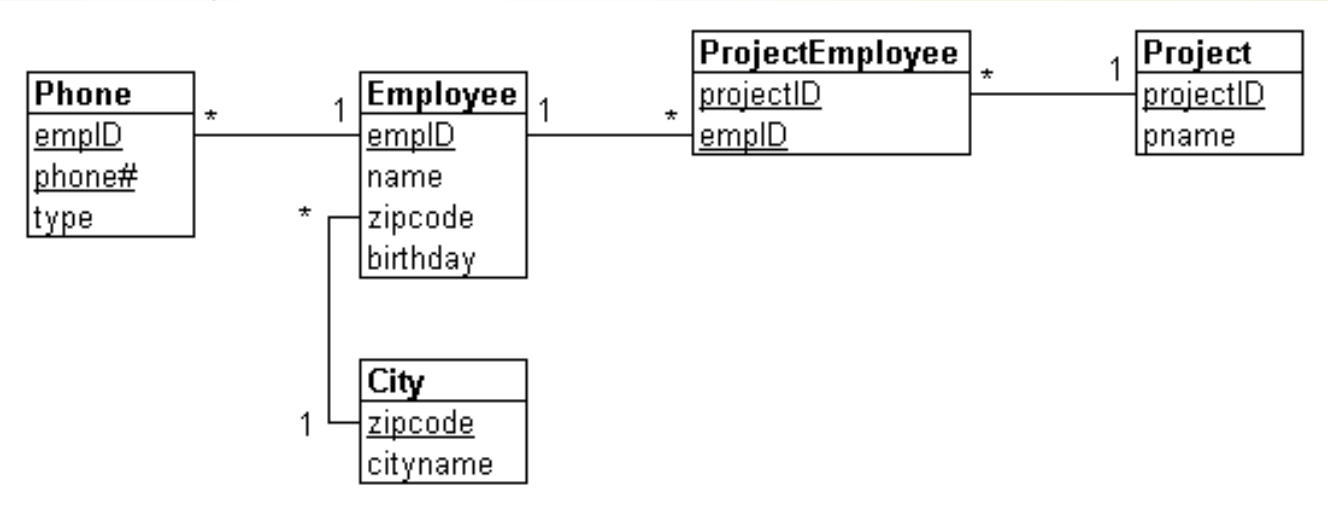


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https://www.youtube.com/watch?v=CZTkgMoqVss

Exercises-4

- Draw the ER model of the database according to the schema given.

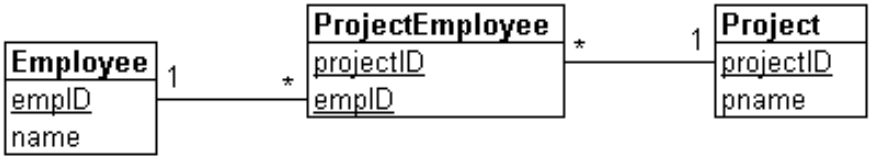


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Exercises-5

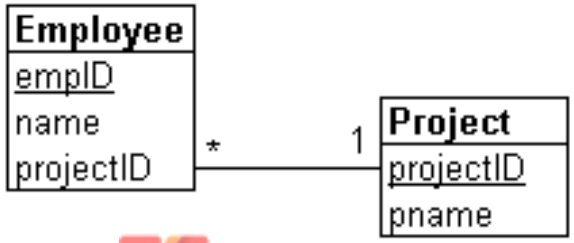
Project Database

In the following design a project may involve many employees and one employee can be assigned to many projects:



Change the design so, that one employee can be assigned to one project only. Still, one project may involve many employees.

Project Database



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Resources

- «Database Management Systems 1», T.ÖZSEVEN, Murathan Publications, 2010
- http://www2.amk.fi/digma.fi/www.amk.fi/opintojaksot/0303011/1142845462205/1142847802793/1142848508953/1142848642251.html
- Carlos Coronel, Steven Morris, and Peter Rob, Database Systems: Design, Implementation, and Management, Cengage Learning.

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